Homework 5

1) Explain how an inverting amplifier differs from a noninverting amplifier.

2)

The output voltage v_o of the circuit of Figure P10.4 is 100 mV with the switch closed. With the switch open, the output voltage is 50 mV. Find the input resistance of the amplifier.

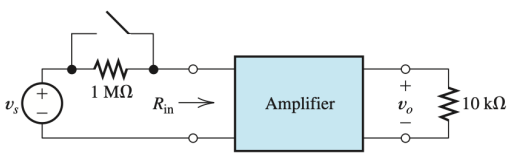

Figure P10.4

3)

A certain amplifier operating with a 100-Ω load has an input resistance of 200 Ω and a power gain of 5000. Determine the current gain and voltage gain of the amplifier.

4)

What is an ideal diode? Draw its volt–ampere characteristic. After solving a circuit with ideal diodes, what check is necessary for diodes initially assumed to be on? Off?

5)

The voltage gain of an amplifier is 5, the load resistance is 100 Ω, and the input resistance of the amplifier is 0.01 MΩ. Determine the current gain and power gain under these conditions.

6)

What is a clipper circuit? Draw an example circuit diagram, including component values, an input waveform, and the corresponding output waveform.

7)

Consider an *npn* silicon transistor that has $\beta = 100$ and $i_B = 0.1$ mA. Sketch i_C versus v_{CE}, for v_{CE} ranging from 0 to 5 V. Repeat for $\beta = 200$.

8)

The diode shown in Figure T9.2 is ideal. Determine the state of the diode and the values of v_x and i_x.

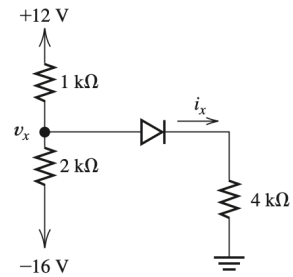

Figure T9.2

9)

Determine the value of β for the transistor of Figure P12.15.

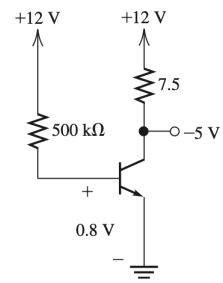

Figure P12.15

10)

Sketch the transfer characteristic (v_o versus v_{in}) to scale for the circuit shown in Figure P9.64. Assume that the diode is ideal.

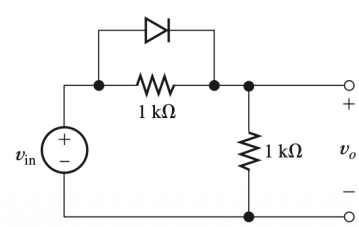

Figure P9.64